



# AI-Powered Smart Traffic Management System

This system combines:

- YOLOv8 Vehicle Detection
- DeepSORT Vehicle Tracking
- Vehicle Speed Estimation
- Traffic Density Analysis
- SQLite Database
- Random Forest Forecasting
- XGBoost Forecasting
- LSTM Forecasting
- Q-Learning Signal Optimization
- Emergency Vehicle Detection



## Traffic Overview

| Average Vehicles | Maximum Vehicles | Minimum Vehicles | Total Records |
|------------------|------------------|------------------|---------------|
| 9.38             | 32               | 1                | 1859          |



## 4-Way Smart Junction

|   | Direction | Vehicles |
|---|-----------|----------|
| 0 | North     |          |
| 1 | South     |          |
| 2 | East      |          |
| 3 | West      |          |

North Traffic : 2

South Traffic : 18

East Traffic : 5

West Traffic : 9



## AI Signal Decision

Decision: North-South Corridor Selected



NORTH SIGNAL : GREEN



SOUTH SIGNAL : GREEN



EAST SIGNAL : RED



WEST SIGNAL : RED

|   | Direction | Signal |
|---|-----------|--------|
| 0 | North     | GREEN  |
| 1 | South     | GREEN  |
| 2 | East      | RED    |
| 3 | West      | RED    |

North-South Traffic : 20

East-West Traffic : 14

Green Signal Time : 40 Seconds



Current Signal Status

 NORTH : OPEN

 SOUTH : OPEN

 EAST : STOP

 WEST : STOP

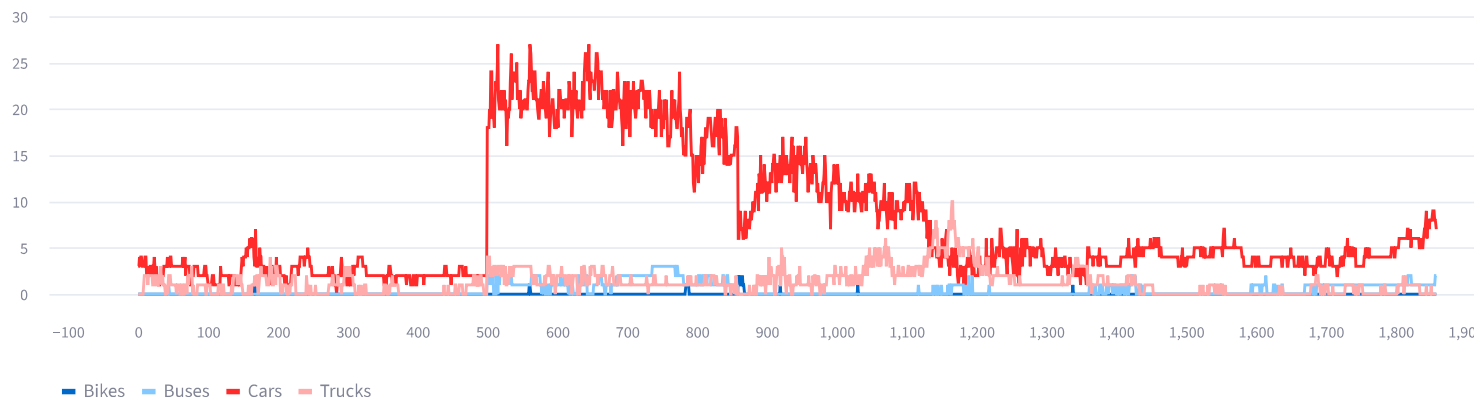


Traffic Dataset

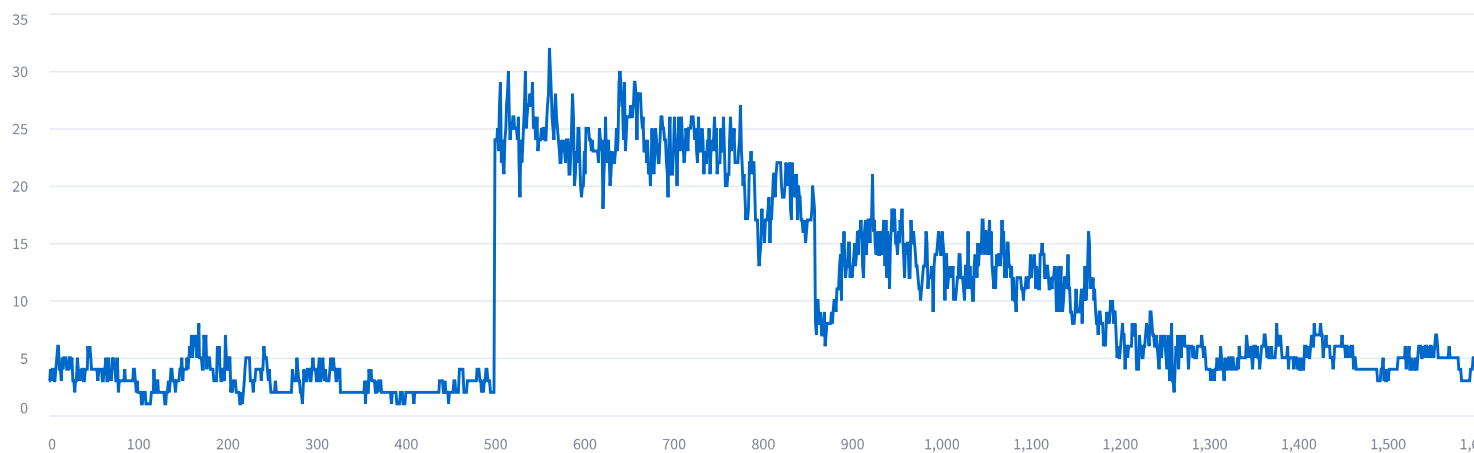
|      | Direction | Cars | Bikes | Buses | Trucks | Total |
|------|-----------|------|-------|-------|--------|-------|
| 1839 | West      | 5    | 0     | 1     | 0      |       |
| 1840 | West      | 6    | 0     | 1     | 0      |       |
| 1841 | West      | 7    | 0     | 1     | 0      |       |
| 1842 | West      | 7    | 0     | 1     | 0      |       |
| 1843 | West      | 8    | 0     | 1     | 0      |       |
| 1844 | West      | 9    | 0     | 1     | 1      |       |
| 1845 | West      | 6    | 0     | 1     | 0      |       |
| 1846 | West      | 8    | 0     | 1     | 0      |       |
| 1847 | West      | 8    | 0     | 1     | 0      |       |
| 1848 | West      | 7    | 0     | 1     | 0      |       |



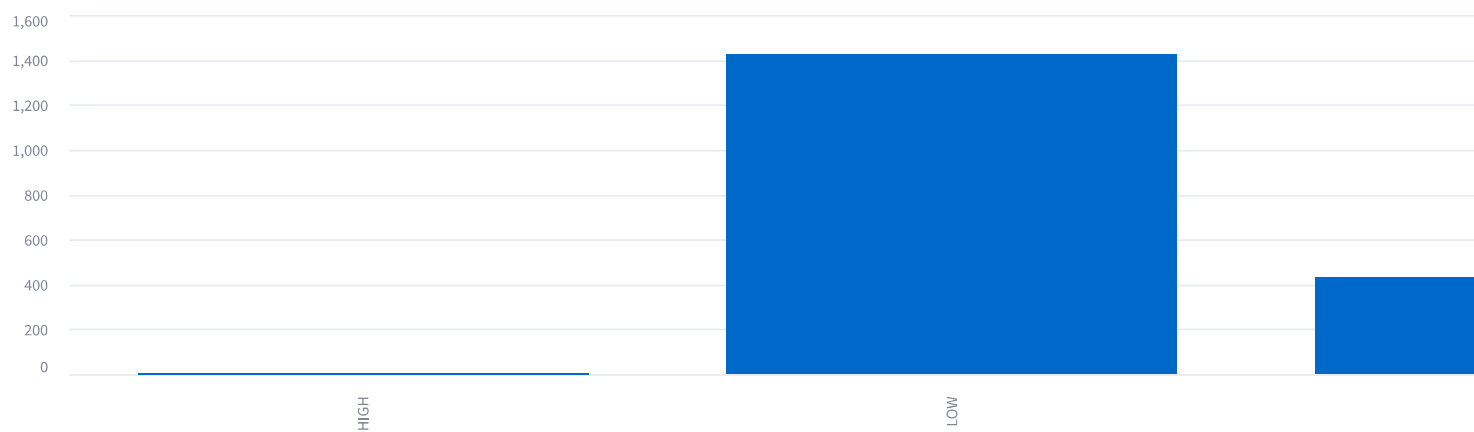
Vehicle Trends



## Total Vehicles



## Density Distribution



## Traffic Analytics

Average Vehicles : 9.38

Maximum Vehicles : 32

Minimum Vehicles : 1

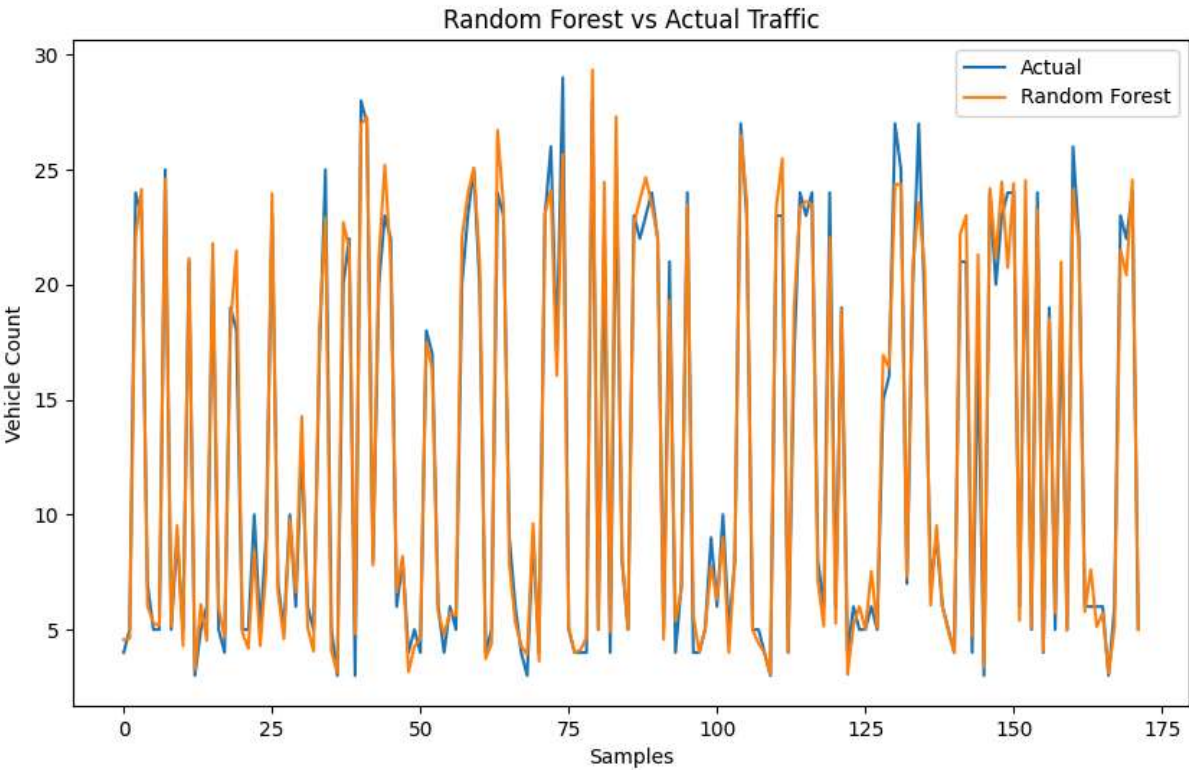
Congestion Level : LOW

📊 Forecasting Models

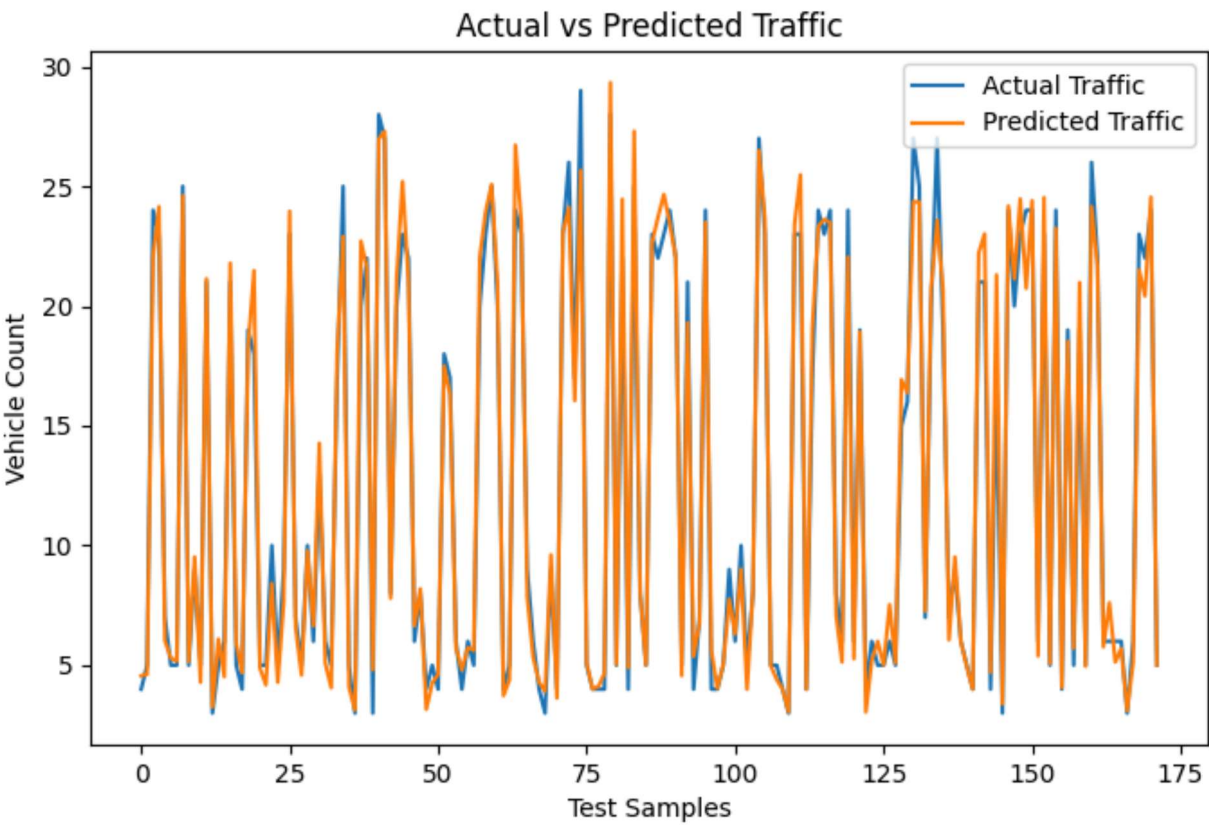
|   | Model         | MAE | RMSE | R2   |
|---|---------------|-----|------|------|
| 0 | Random Forest |     | 0.87 | 1.18 |
| 1 | XGBoost       |     | 1.03 | 1.54 |
| 2 | LSTM          |     | 0.68 | 0.93 |

🏆 Best Model: Random Forest ( $R^2 = 0.981$ )

Model Comparison Graph



Actual vs Predicted Graph



### Database Records

|   | id | vehicle_count | density | avg_speed | sig |
|---|----|---------------|---------|-----------|-----|
| 0 | 1  | 21            | LOW     | 34.5      | Nc  |
| 1 | 2  | 21            | LOW     | 34.5      | Nc  |
| 2 | 3  | 10            | LOW     | 50        | Rc  |
| 3 | 4  | 4             | LOW     | 56        | Rc  |

### Q-Learning Signal Optimization

Q-Learning Agent:

- Monitors traffic density
- Evaluates road congestion
- Selects optimal signal direction
- Minimizes waiting time

### Emergency Vehicle System

Emergency Vehicle Detection Module

If Ambulance Detected: → Priority Signal Activated → Green Corridor Enabled



System Status

|   | Module            | Status    |
|---|-------------------|-----------|
| 0 | YOLO Detection    | Completed |
| 1 | DeepSORT Tracking | Completed |
| 2 | Traffic Analytics | Completed |
| 3 | Random Forest     | Completed |
| 4 | XGBoost           | Completed |
| 5 | LSTM              | Completed |
| 6 | Q-Learning        | Completed |
| 7 | Database          | Completed |
| 8 | 4-Way Junction    | Completed |
| 9 | Dashboard         | Completed |